

CSE46901: Introduction to Robotics

Physical Milestone 2 (can be substituted in place of Sim milestone 2)

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Introduction

Welcome to CSE46901. We have prepared a few milestones before the final project, in which you will use a physical robot to execute a mission given by us. The following milestones will give you an idea how to interact with [ROBOTIS' Turtlebot3 waffle Pi](#) with a [Manipulator Arm](#) using the Robot Operating System.

Submission requirement

Hardware video demonstration submission is not required for milestone2. (i.e. **This milestone is optional and you can replace it with a simulated milestone 2 video.**) However, if you can arrive early in Ulsan for physical classes, you might want to try out physical milestones before the final project. Experience of dealing with physical robots might help your team get ahead in the final project.

We are not trying to differentiate students' performance with milestones. The only grade that will be given is Pass/Fail. However, we do expect you to provide us with evidence that you have followed through the milestones by posting a video of each group following through with the tasks in Part 1 and 2. Requirements of what to be included in the video will be provided below. **Videos of either Simulated milestone2 or Physical milestone2 will get you a pass for the milestone requirement.**

Once your group is done with the video demonstration, please submit it on the blackboard. Each group will submit one video. Rules for robot room usage will apply. Every member will be asked to be present during receiving/handing over the turtlebot in the robot room as our soft-policy to discourage free-riding.

Once we verify that you have submitted the milestone requirement, you will be awarded Pass/Fail. You can resubmit videos to challenge your Fail status until the final project time period so that all groups can eventually pass the milestone requirement with no penalty for resubmission. (i.e. no difference between teams who get it right the first try vs teams who retries). However, if we resume offline lectures, there would be a logistical limit of each team being able to physically interact with the turtlebot.

Instruction

Part1

This tutorial is based on the following tutorial. However, because turtlebot officially does not support ROS Noetic on Ubuntu 20.0.4, do not blindly follow robotis' tutorial. However, you may still head there for more information.

<https://emanual.robotis.com/docs/en/platform/turtlebot3/manipulation/>

This tutorial assumes that you have followed setup progress in milestone 1

Note that if you are using the provided NUC, you do not have sudo authority. To compensate, we provide separate instruction. In the provided NUC all commands requiring sudo are already completed for you.

(If you have followed instruction for setting up your account for NUC, that instruction contains this step so you do not have to do this again)

On your provided NUC:

Your account in the provided NUC has already been set up with all commands that require root. However, you still need to compile few files from source.

1) Set up Open-Manipulator-X

- Compile following pertaining to open manipulator from source

```
cd ~/catkin_ws/src/  
git clone https://github.com/ROBOTIS-GIT/DynamixelSDK.git  
git clone https://github.com/ROBOTIS-GIT/dynamixel-workbench.git  
git clone https://github.com/ROBOTIS-GIT/dynamixel-workbench-msgs.git  
git clone https://github.com/ROBOTIS-GIT/open_manipulator.git  
git clone https://github.com/ROBOTIS-GIT/open_manipulator_msgs.git  
git clone https://github.com/ROBOTIS-GIT/open_manipulator_simulations.git  
git clone https://github.com/ROBOTIS-GIT/robotis_manipulator.git  
cd ~/catkin_ws && catkin_make
```

```
source ~/catkin_ws/devel/setup.bash
```

- Compile following files pertaining to turtlebot3 from source

```
cd ~/catkin_ws/src/  
git clone https://github.com/ROBOTIS-GIT/turtlebot3_manipulation.git
```

```
git clone https://github.com/ROBOTIS-GIT/turtlebot3_manipulation_simulations.git  
cd ~/catkin_ws && catkin_make
```

```
source ~/catkin_ws/devel/setup.bash
```

-Note: **You should not need root access to run catkin_make** for whatever reason.
Make sure you have set up catkin folders correctly. You do not need to worry if you have managed to complete milestone 1.

(Instruction same as Simulated Milestone 2's remote pc setup)

If you are using your own computer

On your own Computer:

1) Set up Open-Manipulator-X:

- Use a root account to install a few more ROS packages according to
https://emanual.robotis.com/docs/en/platform/openmanipulator_x/ros_setup/#install-ubuntu-on-pc

```
sudo apt-get update && sudo apt-get upgrade
```

```
sudo apt-get install ros-noetic-ros-controllers ros-noetic-gazebo*  
ros-noetic-moveit*
```

```
sudo apt-get install ros-noetic-industrial-*
```

- Compile following pertaining to open manipulator from source

```
cd ~/catkin_ws/src/  
git clone https://github.com/ROBOTIS-GIT/DynamixelSDK.git  
git clone https://github.com/ROBOTIS-GIT/dynamixel-workbench.git  
git clone https://github.com/ROBOTIS-GIT/dynamixel-workbench-msgs.git  
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git clone https://github.com/ROBOTIS-GIT/open_manipulator_msgs.git  
git clone https://github.com/ROBOTIS-GIT/open_manipulator_simulations.git  
git clone https://github.com/ROBOTIS-GIT/robotis_manipulator.git  
cd ~/catkin_ws && catkin_make
```

```
source ~/catkin_ws/devel/setup.bash
```

- Compile following files pertaining to turtlebot3 from source

```
cd ~/catkin_ws/src/  
git clone https://github.com/ROBOTIS-GIT/turtlebot3_manipulation.git  
git clone  
https://github.com/ROBOTIS-GIT/turtlebot3\_manipulation\_simulations.git  
cd ~/catkin_ws && catkin_make
```

```
source ~/catkin_ws/devel/setup.bash
```

Part2

Next we need you to demonstrate your simulated environment with the operation of a robotic manipulator on Gazebo. Two ways will be presented here: rviz and Robotis GUI.

How to run simulation using Rviz

Warning: Manipulator arm on Waffle_pi is strong enough to even lift Waffle_pi itself. Do not command extreme movements that might damage motors on the manipulator arm

Warning: **Powering off the opencr board will drop the arm instantly. Make sure you hold the arm before powering off turtlebot3.**

Warning: Motor wires on the manipulator arm might interfere with the spinning lidar sensor when commanded to be in a certain position. **Take care not to damage the wires on the manipulator arm.**

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<https://emanual.robotis.com/docs/en/platform/turtlebot3/manipulation/>

This tutorial assumes that you have completed the setup process for remote-pc correctly on milestone 1 and prepared raspberry pi3's sd cards properly based on Physical milestone 1. We completed that step and are providing you with the preflashed sd cards. However, if you suspect your sd card is corrupted, you can attempt to reflash the sd card with the preconfigured sd image we are providing you or can start over using instructions from Physical milestone 1.

1) Make sure ip address setting on bashrc is still correct by typing

```
ifconfig
```

and

```
vi ~/.bashrc
```

If you changed any of the settings, type

```
source ~/.bashrc
```

2) Test Open-Manipulator-X (Moveit RViz):

- On Remote-PC

```
roscore
```

- On TurtleBot (via ssh or directly)

```
roslaunch turtlebot3_bringup turtlebot3_robot.launch
```

- On Remote-PC, open a new terminal and launch (each in separate terminal windows)

```
roslaunch turtlebot3_manipulation_bringup turtlebot3_manipulation_bringup.launch
```

```
roslaunch turtlebot3_manipulation_moveit_config move_group.launch
```

```
roslaunch turtlebot3_manipulation_moveit_config moveit_rviz.launch
```

- You can now move the blue ball around and click "Plan and Execute" to move the arm to desired location.

- Try out different things within Rviz gui to get a feel of turtlebot3 with manipulator arm's capabilities.

-Note: Under Motion planning -> Planning-> Commands box there exists a feedback message that give feedback if turtlebot managed to successfully execute your commands. If output is "fail" or for whatever reason the arm does not move according to your command, check if "turtlebot3_manipulation_bringup" is still running. Some ros nodes essential for operation of the manipulator arm can crash without your intervention.

How to run simulation using ROBOTIS GUI

Warning: Manipulator arm on Waffle_pi is strong enough to even lift Waffle_pi itself. Do not command extreme movements that might damage motors on the manipulator arm

Warning: **Powering off the opencr board while the arm is not in default position will drop the arm instantly. Make sure you hold the arm before powering off turtlebot3.**

Warning: Motor wires on the manipulator arm might interfere with the spinning lidar sensor when commanded to be in a certain position. **Take care not to damage the wires on the manipulator arm.**

This tutorial is based on the following tutorial. However, because turtlebot officially does not support ROS Noetic on ubuntu 20.0.4, do not blindly follow robotis' tutorial. However, you may still head there for more information.

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- On Remote-PC, open a new terminal and launch (each in separate terminal windows)

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roslaunch turtlebot3_manipulation_bringup turtlebot3_manipulation_bringup.launch
```

```
roslaunch turtlebot3_manipulation_moveit_config move_group.launch
```

```
roslaunch turtlebot3_manipulation_gui turtlebot3_manipulation_gui.launch
```

- Click "Timer Start" on the GUI
- Click "Read joint angle" on the GUI under joint space
- You can now issue commands by changing Joint 1 ~ Joint 4's angles and clicking "send" button or manipulate gripper by clicking "Gripper open" button or "Gripper close" button.
- Try out different things within the Open manipulator control gui to get a feel of turtlebot3 with manipulator arm's capabilities.

-Note: If the arm does not move according to your command, check if "turtlebot3_manipulation_bringup" is still running. Some ros nodes essential for operation of the manipulator arm can crash without your intervention.

3) After testing, move on to test-manipulation-rviz.txt

Demo requirement

You will have to submit some photos and/or videos showing that you have successfully controlled the manipulator via

(1) rviz and

(2) the ROBOTIS GUI Controller.

Please try to learn the user interface of rviz and the ROBOTIS GUI Controller, and demonstrate what you have learnt in the videos.

Topic explanation is not required for physical milestones.